



INTRODUCTION TO ANSIBLE PART 2

ANSIBLE VAULT, ROLES AND JINJA2 TEMPLATES

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AGENDA

- **WHAT IS ANSIBLE VAULT?**
- **INTRODUCTION TO JINJA2 TEMPLATES**
- **JINJA2 TEMPLATES EXAMPLE**
- **WHAT ARE ANSIBLE ROLES?**
- **COMPONENTS OF ANSIBLE ROLES**
- **WHAT IS ANSIBLE GALAXY?**
- **DEMO: CREATING AN ANSIBLE ROLE**

WHAT IS ANSIBLE VAULT?

- A mechanism for encrypting sensitive information within your Ansible code
- Used for hiding the content of variables, files, templates and more
- Ansible reads the secrets at runtime if provided with the vault password
- Commands:
 - `ansible-vault create`
 - `ansible-vault encrypt`
 - `ansible-vault decrypt`
 - `ansible-vault edit`
 - `ansible-vault view`
 - ...



INTRODUCTION TO JINJA2 TEMPLATES

- Files that have access to playbook variables
- Can replace values in files dynamically
- Utilizes for loops, if conditions...
- Used by the template module
- Processed on Ansible master and then sent to target hosts



JINJA2 TEMPLATES EXAMPLE

```
1 [Unit]
2 Description={{ app_description }}
3 After=network.target
4
5 [Service]
6 {% if service_type == 'forking' %}
7 Type=forking
8 {% elif service_type == 'oneshot' %}
9 Type=oneshot
10 {% else %}
11 Type=simple
12 {% endif %}
13 WorkingDirectory={{ app_install_path }}/my_app/
14 {% for env_var in environment_variables %}
15 Environment={{ env_var.name|upper }}={{ env_var.value }}
16 {% endfor %}
17 ExecStart={{ app_install_path }}/my_app/my_app
18 ExecStop=/bin/kill $MAINPID
19 Restart=on-failure
20
21 [Install]
22 WantedBy=multi-user.target
```

Variable interpolation, loops and conditions in a systemd service Jinja2 template

```
1 ### Variable file ###
2
3 app_description: My App
4 service_type: simple
5 app_install_path: /opt/apps
6 environment_variables:
7   - name: config_file
8     value: conf.yml
9   - name: application_port
10     value: 8080
11
```

Variables used in the systemd Jinja2 template

```
1 [Unit]
2 Description=My App
3 After=network.target
4
5 [Service]
6 Type=simple
7 WorkingDirectory=/opt/apps/my_app/
8 Environment=CONFIG_FILE=conf.yml
9 Environment=APPLICATION_PORT=8080
10 ExecStart=/opt/apps/my_app/my_app
11 ExecStop=/bin/kill $MAINPID
12 Restart=on-failure
13
14 [Install]
15 WantedBy=multi-user.target
```

Templated systemd service file

```
tasks:
  - name: Template systemd service file
    become: yes
    template:
      src: my_app.service.j2
      dest: /etc/systemd/system/my_app.service
      mode: 0644
    notify:
      - daemon_reload
      - start service
```

Ansible task utilizing the template module

WHAT ARE ANSIBLE ROLES?

- Independent components which allow reuse of common configuration steps
- All Ansible roles have a standard folder structure, organizing their components into subdirectories
- Roles must be imported from within a playbook and run as part of it



ANSIBLE

COMPONENTS OF ANSIBLE ROLES

- **README.md** - description of the role
- **defaults/** - default values for the role variables
- **files/** - static files used by the role
- **handlers/** - list handlers used by the role (same as a handlers list in a playbook)
- **meta/** - role metadata (dependencies, author, company, license...)
- **tasks/** - list of tasks executed by the role (same as a task list in a playbook)
- **templates/** - Jinja2 template files used by the role
- **tests/** - role tests
- **vars/** - other variables for the role

```
example-role
├── README.md
├── defaults
│   └── main.yml
├── files
│   └── example-file.conf
├── handlers
│   └── main.yml
├── meta
│   └── main.yml
├── tasks
│   └── main.yml
├── templates
│   └── example-template.json.j2
├── tests
│   ├── inventory
│   └── test.yml
└── vars
    └── main.yml
```

WHAT IS ANSIBLE GALAXY?

- A repository for Ansible collections and roles
- Massive open-source community





DEMO: CREATING AN ANSIBLE ROLE

ANSIBLE LEARNING RESOURCES

ANSIBLE DOCUMENTATION

- <https://docs.ansible.com/ansible/latest/index.html>

JINJA2 TEMPLATES

- <https://jinja2docs.readthedocs.io/en/stable/templates.html>

LIST OF AVAILABLE MODULES

- [List of modules](#)

ANSIBLE GALAXY

- <https://galaxy.ansible.com/>

THANK YOU!

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